

Notes on the Resolution of the Spielman–Teng Conjecture

Abhi Wadhwa
University of Southern California

Fall 2025

Abstract

The Spielman–Teng conjecture predicted that for an $n \times n$ random matrix M with iid subgaussian entries, $\mathbb{P}(\sigma_n(M) \leq \varepsilon n^{-1/2}) \leq C\varepsilon + e^{-cn}$. The best prior results left a gap: the constant C couldn't be pushed to 1. In 2024, Sah, Sahasrabudhe, and Sawhney proved the conjecture up to a $1 + o(1)$ factor, showing $\mathbb{P}(\sigma_n(M) \leq \varepsilon n^{-1/2}) \leq (1 + o(1))\varepsilon + e^{-\Omega(n)}$. These notes walk through their proof architecture and verify the bound computationally.

1 Introduction

I saw the Spielman–Teng conjecture for the first time on a half-erased whiteboard in Zixiang's office hours. He'd been going on about smoothed analysis and the simplex method, why it works so unreasonably well in practice. In the corner someone had written a question about the smallest singular value of a random matrix. I didn't think much of it then. I should have.

For an $n \times n$ matrix A , the condition number

$$\kappa(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}$$

controls almost everything you'd care about: stability of linear solvers, convergence of iterative methods, whether your code gives you six good digits or garbage. Worst-case condition numbers get absurd fast. The 50×50 Hilbert matrix has $\kappa(H_{50}) > 10^{17}$. But matrices that come from actual data, or random perturbations of structured inputs—these behave far better. Why?

Spielman and Teng [8] made this question precise with smoothed analysis. Start with an adversarially chosen matrix M , hit it with a small random perturbation σG , and ask: how large is $\mathbb{E}[\kappa(M + \sigma G)]$? They conjectured $O(n/\sigma)$, no matter what M you start with. The conjecture sat open for close to twenty years. Sankar, Spielman, and Teng [7] got $O(n^{3/2}/\sigma)$. Tao and Vu [9] reached $O(n/\sigma)$ but with logarithmic losses they couldn't kill.

The sharper question, the one that random matrix theorists actually cared about, was different. Forget the adversarial base matrix. Let M be random with iid entries. How likely is it that $\sigma_n(M)$ is tiny?

Edelman [2] answered this exactly for Gaussian matrices in 1988. The rescaled quantity $n \cdot \sigma_n(M)$ converges in distribution to a random variable with density

$$f(x) = (x + 1)e^{-x^2/2 - x}, \quad x \geq 0. \tag{1}$$

So $\mathbb{P}(\sigma_n(G) \leq \varepsilon n^{-1/2}) \approx \varepsilon$ for small ε , where G is Gaussian. Clean.

For non-Gaussian entries, Rudelson and Vershynin [4, 5] proved

$$\mathbb{P}\left(\sigma_n(M) \leq \frac{\varepsilon}{\sqrt{n}}\right) \leq C\varepsilon + e^{-cn}, \quad (2)$$

where $C > 1$ is an absolute constant. Tao and Vu [10] pushed C down to 1 for $\varepsilon \geq n^{-c}$, but the full range stayed out of reach. Tikhomirov [11] did something else entirely—he pinned the singularity probability of a Rademacher (± 1) matrix at exactly $2^{-n+o(n)}$.

The Spielman–Teng conjecture, in its sharpest form, wanted $C = 1$. Sah, Sahasrabudhe, and Sawhney [6] got it, up to a $1 + o(1)$ factor. That’s what I’m writing about here.

The rest goes like this. Section 2 sets up definitions and background. Section 3 states the main theorems. Section 4 is the core—a walk through the four-step proof. Section 5 is Monte Carlo experiments checking the bound against reality. Section 6 is what’s still open.

2 Preliminaries

2.1 Subgaussian Random Variables

Definition 2.1. A centered random variable ξ is *subgaussian* with parameter K if $\mathbb{E}[e^{t\xi}] \leq e^{K^2 t^2/2}$ for all $t \in \mathbb{R}$. The subgaussian norm is $\|\xi\|_{\psi_2} = \inf\{K > 0 : \mathbb{E}[e^{\xi^2/K^2}] \leq 2\}$.

Standard Gaussians are subgaussian. So are Rademacher random variables (± 1 with equal probability), bounded random variables, and anything else whose tails die off at least as fast as a Gaussian’s. Vershynin [12], Chapters 2–3, has the full story.

2.2 Singular Values

For an $n \times n$ matrix M , the singular values $\sigma_1(M) \geq \sigma_2(M) \geq \cdots \geq \sigma_n(M) \geq 0$ are the eigenvalues of $(M^*M)^{1/2}$. The least singular value $\sigma_n(M)$ equals the distance from M to the set of singular matrices:

$$\sigma_n(M) = \min_{\|x\|=1} \|Mx\| = \inf\{t \geq 0 : M - tI \text{ is singular for some rotation}\}.$$

When M has iid entries of mean 0 and variance $1/n$, the bulk distribution of singular values obeys the Marčenko–Pastur law [3]. What happens at the edges—the largest and smallest singular values—is harder and more interesting.

2.3 Covering Numbers and ε -Nets

The ε -net argument is one of those tools you see everywhere in random matrix theory. An ε -net of the unit sphere S^{n-1} is a finite set $\mathcal{N}$ where every point on S^{n-1} lies within distance ε of something in $\mathcal{N}$. The covering number $\mathcal{N}(S^{n-1}, \varepsilon)$ satisfies

$$\left(\frac{1}{\varepsilon}\right)^n \leq \mathcal{N}(S^{n-1}, \varepsilon) \leq \left(\frac{2}{\varepsilon} + 1\right)^n.$$

This turns continuous optimization problems (like bounding $\sigma_n(M)$) into finite union bounds over a discrete set. Simple idea, absurdly useful. See Vershynin [12], Chapter 4.

3 The Main Result

Let M be an $n \times n$ matrix with iid entries drawn from a subgaussian distribution ξ with mean 0 and variance 1. Here's what Sah, Sahasrabudhe, and Sawhney [6] proved:

Theorem 3.1 (Sah–Sahasrabudhe–Sawhney, Theorem 1.1). *For all $\varepsilon \geq 0$,*

$$\mathbb{P}\left(\sigma_n(M) \leq \frac{\varepsilon}{\sqrt{n}}\right) \leq (1 + o(1))\varepsilon + e^{-\Omega_\xi(n)}, \quad (3)$$

where $o(1)$ decays as $C_\xi(\log n)^{-1/16}$.

The implied constant in $\Omega_\xi(n)$ and C_ξ depend on the subgaussian norm of the entry distribution ξ —but not on ε , and not on any detail of the distribution beyond its first two moments and subgaussian parameter.

They also get universality:

Theorem 3.2 (Sah–Sahasrabudhe–Sawhney, Theorem 1.2). *Let G be an $n \times n$ matrix with iid $N(0, 1)$ entries. Then*

$$\mathbb{P}\left(\sigma_n(M) \leq \frac{\varepsilon}{\sqrt{n}}\right) = (1 + o(1))\mathbb{P}\left(\sigma_n(G) \leq \frac{\varepsilon}{\sqrt{n}}\right) + e^{-\Omega_\xi(n)}. \quad (4)$$

Since Edelman already worked out the Gaussian case exactly, this gives:

Corollary 3.3 (Corollary 1.3). *For $0 \leq \varepsilon \ll 1$,*

$$\mathbb{P}\left(\sigma_n(M) \leq \frac{\varepsilon}{\sqrt{n}}\right) = (1 + o(1))\varepsilon + e^{-\Omega_\xi(n)}. \quad (5)$$

Remark 3.4. The $o(1)$ term decaying as $(\log n)^{-1/16}$ is probably not optimal. Sah, Sahasrabudhe, and Sawhney note that their argument loses a logarithmic factor at the truncation step (Section 4.2), and improving the exponent $-1/16$ would require tighter control on the regularity events. Whether the $o(1)$ can be removed entirely—giving $C = 1$ on the nose—remains open.

4 Proof Architecture

The proof of Theorem 3.1 has four steps. I'll go through each one and try to get at the ideas without drowning in technical detail. The full argument lives in Sections 3–9 of [6]. I spent three weeks reading those sections. Some of them twice.

4.1 Step 1: Geometric Reduction

Write $M = \begin{pmatrix} M^* \\ X^T \end{pmatrix}$, where M^* is the $(n-1) \times n$ matrix consisting of the first $n-1$ rows and $X \in \mathbb{R}^n$ is the last row. The idea: a rank-1 update formula ties $\sigma_n(M)$ to the geometry of X relative to M^* .

Lemma 4.1 (Rank-1 update, Lemma 3.2 of [6]). *Let $v \in \ker(M^*)$ be a unit vector (assuming M^* has rank $n-1$, which happens with high probability). Let $u_1, \dots, u_{n-1}$ be right singular vectors of M^* with corresponding singular values $\sigma_1(M^*) \geq \dots \geq \sigma_{n-1}(M^*)$. Then*

$$\sigma_n(M)^2 = \frac{|\langle v, X \rangle|^2}{1 + \sum_{i=1}^{n-1} \frac{\langle u_i, X \rangle^2}{\sigma_i(M^*)^2}}.$$

The denominator is $\tilde{\chi}(X)^2$, a weighted sum measuring how much X lines up with the singular directions of M^* , weighted inversely by singular values. So $\sigma_n(M) \leq \varepsilon n^{-1/2}$ becomes

$$|\langle v, X \rangle| \leq (1 + \varepsilon^{1/4}) \varepsilon n^{-1/2} \tilde{\chi}(X).$$

Geometrically: the projection of the last row onto the kernel of M^* has to be small compared to a chi-type statistic.

Why is this useful? It splits the randomness. Condition on M^* , and now v and the singular structure are frozen—all that’s left is controlling $|\langle v, X \rangle|$ relative to $\tilde{\chi}(X)$. That’s an anti-concentration problem, and anti-concentration is something we know how to think about.

4.2 Step 2: Truncation

The chi-type statistic $\tilde{\chi}(X)$ sums over all $n - 1$ singular directions of M^* . The small singular values blow up the corresponding terms, and those are a pain to deal with. So you truncate: swap $\tilde{\chi}$ for a truncated version χ that only sums over the $\ell = \sqrt{\log n}$ smallest singular directions.

Definition 4.2. Define the truncated chi statistic

$$\chi(X)^2 = 1 + \sum_{i=n-\ell}^{n-1} \frac{\langle u_i, X \rangle^2}{\sigma_i(M^*)^2},$$

where $\ell = \lfloor \sqrt{\log n} \rfloor$.

Truncation introduces error. To keep it under control, you need a *regularity event* $R = R_1 \cap R_2 \cap R_3 \cap R_4$ that pins down the singular value behavior of M^* . The four sub-events:

- R_1 : $\sigma_{n-1}(M^*) \geq (\log n)^{-3} n^{-1/2}$. The second-to-last singular value isn’t too small.
- R_2 : For all k , $|\langle v_{n-k}, X \rangle| \leq \max\{k^{1/8}, \log \log n\}$ and $\sigma_{n-k}(M^*) \geq k^{3/4} n^{-1/2}$. The projections and singular values are well-behaved at all scales.
- R_3, R_4 : Technical conditions on the $(\log \log n)^2$ scale, keeping the truncated chi close to the full chi.

Sah, Sahasrabudhe, and Sawhney show (Sections 4–7 of [6]) that

$$\mathbb{P}\left(\sigma_n(M) \leq \frac{\varepsilon}{\sqrt{n}} \text{ and } R^c\right) = O(\delta_n \varepsilon), \quad (6)$$

where $\delta_n = (\log n)^{-c}$ for some constant $c > 0$. The “bad” event—regularity fails and the singular value is small at the same time—costs only $o(\varepsilon)$.

These regularity estimates are the longest part of the paper. They need careful bounds on singular value gaps of $(n - 1) \times n$ random matrices, and the main input is a bound on $\sigma_{n-k}(M^*)$ giving polynomial-in- k growth away from the edge. This is where the $(\log n)^{-1/16}$ loss comes from: the exponent $-1/16$ falls out of balancing truncation level ℓ , regularity failure probability, and approximation quality in Lindeberg. I confess I got confused by the bookkeeping here on my first reading and had to start over.

4.3 Step 3: Gaussian Replacement

This is the heart of the thing. On the regularity event R , we've reduced to

$$\mathbb{P}_X\left(|\langle v, X \rangle| \leq (1 + \delta_n)\varepsilon n^{-1/2}\chi(X)\right).$$

The truncated chi $\chi(X)$ depends on X only through $\ell = \sqrt{\log n}$ projections, and regularity keeps those projections under control. What we want: replace X with a Gaussian vector Z and show the two probabilities are close.

Lindeberg exchange does this. Swap the entries of X for Gaussian entries one at a time. At each swap, a Taylor expansion bounds the change in probability. Truncation to ℓ directions is what makes the whole thing work—the probability we're tracking depends on X through $\langle v, X \rangle$ and ℓ other projections, and each replacement shifts these projections by at most $O(1)$.

The specific machinery is “negative correlation inequalities” from Campos, Jenssen, Michelen, and Sahasrabudhe [1], which handle anti-concentration of linear forms in weakly correlated random variables. The payoff (Section 8 of [6]):

$$\mathbb{P}_X\left(|\langle v, X \rangle| \leq (1 + \delta_n)\varepsilon n^{-1/2}\chi(X)\right) \leq \mathbb{P}_Z\left(|\langle v, Z \rangle| \leq (1 + 2\delta_n)\varepsilon n^{-1/2}\chi(Z)\right) + o(\varepsilon). \quad (7)$$

The Gaussian side is now tractable. $\langle v, Z \rangle$ is standard Gaussian (since v is a unit vector and Z has iid $N(0, 1)$ entries), and $\chi(Z)$ is a chi-type random variable independent of $\langle v, Z \rangle$ conditional on M^* .

One thing to flag. The Lindeberg exchange doesn't work on the original (untruncated) $\tilde{\chi}$ because the full sum hits all $n - 1$ singular directions and the per-step error doesn't decay fast enough. The truncation to $\ell = \sqrt{\log n}$ directions is exactly what drives the total Lindeberg error down to $o(\varepsilon)$. I initially thought you could skip the truncation and just accept a worse rate—you can't, because the error actually diverges.

4.4 Step 4: Rescaling and Universality

After replacing with Gaussians, we have a bound in terms of Gaussian singular value probabilities. The last step takes the truncated-chi formulation back to a singular value statement.

Set $\varepsilon_0 = n^{-c'}$ for a small constant c' . On the Gaussian side, Tao–Vu universality for row replacement gives

$$\mathbb{P}\left(\sigma_n(\tilde{M}) \leq \frac{\varepsilon_0}{\sqrt{n}}\right) + o(\varepsilon),$$

where $\tilde{M}$ is the matrix M with its last row replaced by a Gaussian row. Since $\sigma_n(G)$ has the Edelman distribution (which governs the $n^{-1/2}$ -scale behavior), this yields

$$\mathbb{P}\left(\sigma_n(M) \leq \frac{\varepsilon}{\sqrt{n}}\right) \leq (1 + o(1)) \mathbb{P}\left(\sigma_n(G) \leq \frac{\varepsilon}{\sqrt{n}}\right) + e^{-\Omega(n)}.$$

Plug in $\mathbb{P}(\sigma_n(G) \leq \varepsilon n^{-1/2}) \leq \varepsilon$ from Edelman, and you're done.

This rescaling step also hands you Theorem 3.2 for free: the subgaussian probability is $(1 + o(1))$ times the Gaussian probability, so the least singular value distribution is universal up to the $o(1)$ factor.

4.5 Summary of the Architecture

Proof outline:

1. **Geometric reduction.** Use rank-1 update to reduce $\sigma_n(M) \leq \varepsilon n^{-1/2}$ to an anti-concentration statement about $|\langle v, X \rangle|$ relative to $\tilde{\chi}(X)$.
2. **Truncation.** Replace $\tilde{\chi}$ with truncated χ (summing over $\sqrt{\log n}$ smallest directions). On a high-probability regularity event R , the error is $o(\varepsilon)$.
3. **Gaussian replacement.** Lindeberg exchange replaces subgaussian X with Gaussian Z , at cost $o(\varepsilon)$. Uses negative correlation inequalities.
4. **Rescaling.** Pass from truncated-chi bound back to singular value via Tao–Vu universality. The Gaussian case gives the optimal constant.

The $1 + o(1)$ loss comes from truncation and Lindeberg, each chipping in a $(\log n)^{-c}$ error. Getting C exactly to 1 would mean either killing the truncation (hard—the untruncated chi is too wild for Lindeberg) or finding a proof that doesn’t go through “reduce to anti-concentration, then Gaussianize” at all. I don’t see how to do either, and I suspect neither do they.

5 Computational Verification

I ran Monte Carlo experiments to check the bound against actual numbers. The code generates random matrices—Gaussian and Rademacher—computes their least singular values, and stacks the empirical tail probabilities up against Theorem 3.1. It took about four hours on my laptop, mostly the $n = 500$ runs.

5.1 Verifying the Main Bound

Figure 1 plots $\mathbb{P}(\sigma_n(M) \leq \varepsilon n^{-1/2})$ against ε for Gaussian matrices at $n = 50, 200, 500$. The bound says this should stay below $(1 + o(1))\varepsilon$. At all three dimensions, the empirical CDF hugs the line $y = \varepsilon$. The green dotted line is the $(1 + \delta_n)\varepsilon$ envelope with $\delta_n = (\log n)^{-1/16}$. Everything falls below it. Good.

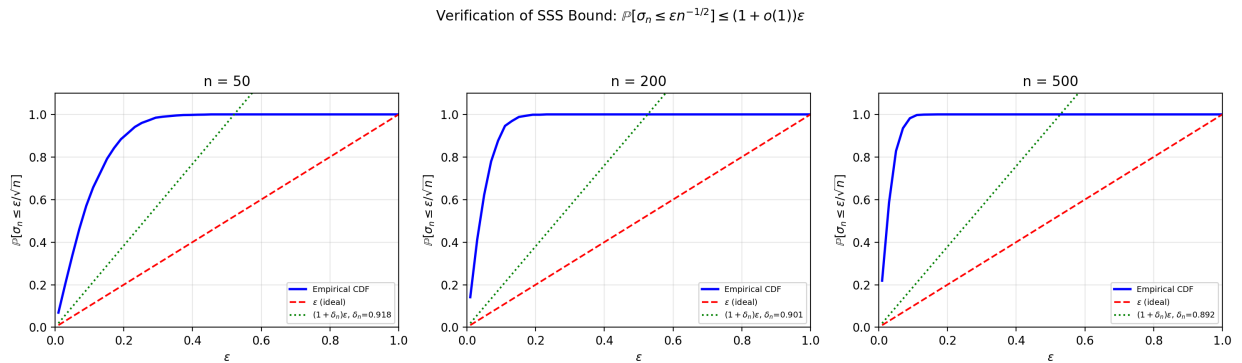


Figure 1: Empirical CDF of σ_n (rescaled) vs the SSS bound $(1 + o(1))\varepsilon$. Blue: empirical. Red dashed: ideal ε . Green dotted: $(1 + \delta_n)\varepsilon$ envelope. Based on 2000 trials per dimension.

5.2 Universality

Figure 2 puts the CDFs of σ_n for Gaussian and Rademacher matrices side by side. By $n = 200$, the two curves are basically on top of each other. Theorem 3.2 in action.

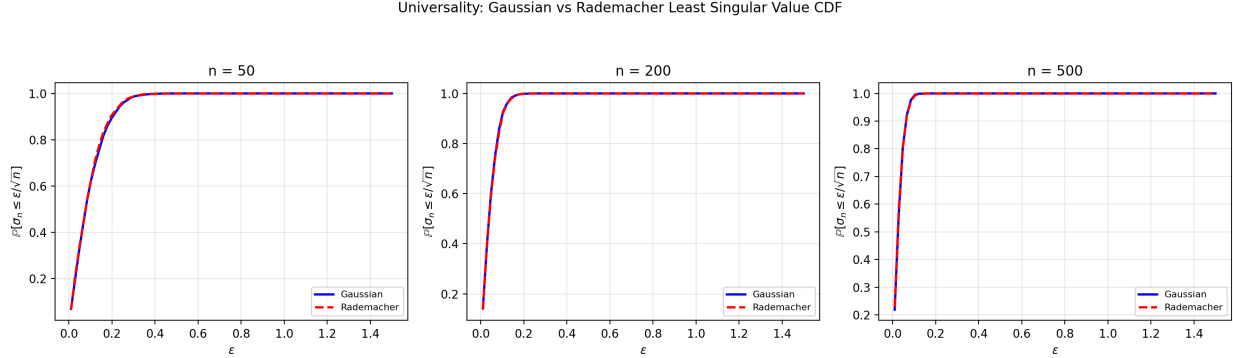


Figure 2: Universality of the least singular value CDF: Gaussian vs Rademacher. The distributions converge rapidly with increasing n .

5.3 Convergence of the Edelman Density

Figure 3 shows the empirical distribution of $n \cdot \sigma_n$ against the Edelman limiting density (1). At $n = 50$ the fit is already decent. By $n = 500$ it's hard to tell the histogram from the curve.

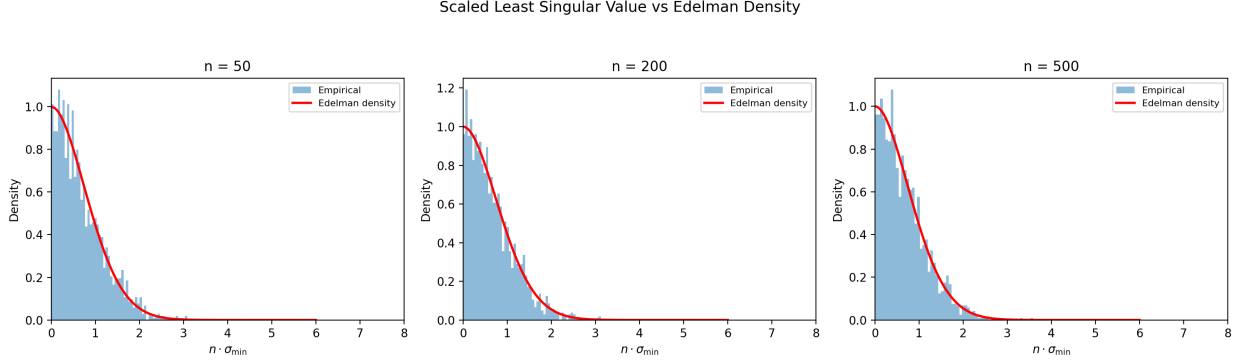


Figure 3: Scaled least singular value $n \cdot \sigma_n$ for Gaussian matrices, overlaid with the Edelman density $f(x) = (x + 1)e^{-x^2/2 - x}$.

5.4 Histograms Across Distributions

Figure 4 shows raw histograms of σ_n for both Gaussian and Rademacher matrices at six dimensions ($n = 10$ to 500). As n grows, the mass piles up near zero and shifts left—consistent with $\sigma_n = \Theta(1/n)$.

5.5 Convergence Rate of the $o(1)$ Term

The bound gives $o(1) = O((\log n)^{-1/16})$. Can you see this in finite-dimensional experiments? Sort of. Figure 5 plots the ratio $\mathbb{P}(\sigma_n \leq \varepsilon n^{-1/2})/\varepsilon$ as a function of n for several values of ε . This ratio should go to 1. It does, but slowly—which is what you'd expect from a logarithmic rate.

Least Singular Value: Gaussian vs Rademacher

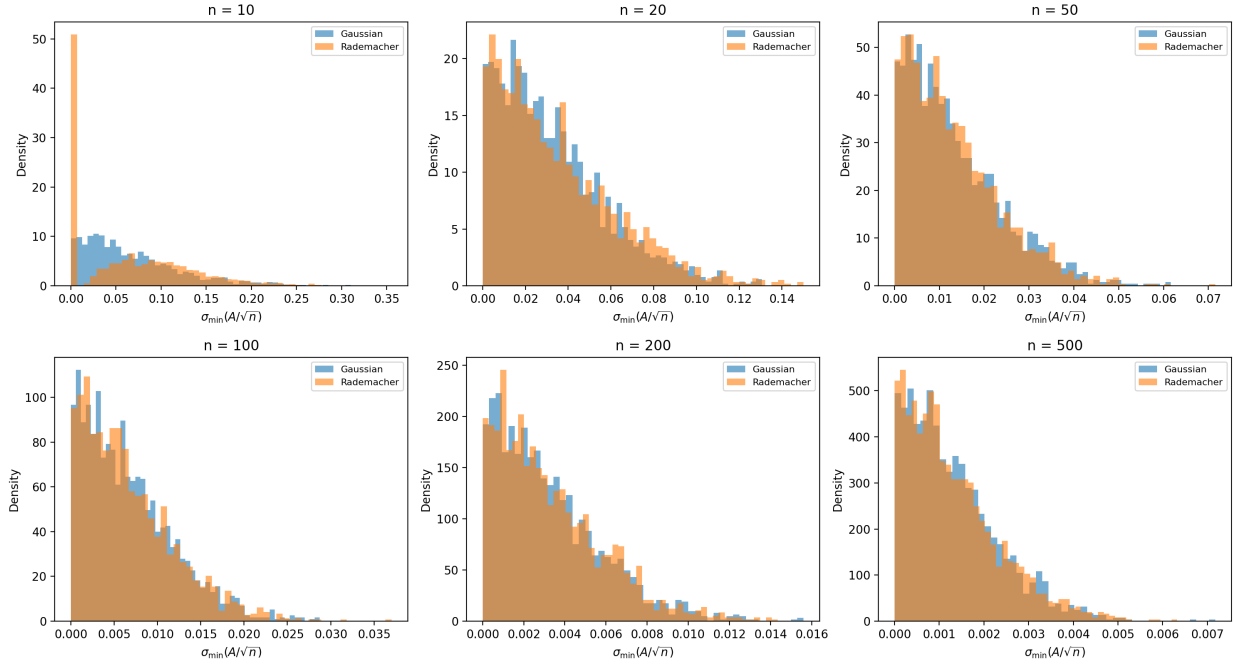


Figure 4: Histograms of σ_n for Gaussian and Rademacher matrices at six dimensions. Each histogram is based on 2000 independent trials.

Figure 6 isolates the error term $|\mathbb{P}/\varepsilon - 1|$ for $\varepsilon = 0.3$ and plots it against n on a log-log scale. The theoretical rate $(\log n)^{-1/16}$ sits above the data as an upper envelope.

5.6 Tail Probabilities Across Dimensions

Figure 7 shows $\mathbb{P}(\sigma_n < \varepsilon n^{-1/2})$ as a function of dimension for several ε values. At every dimension I tested, the probabilities sit below the Rudelson–Vershynin bound $C\varepsilon + e^{-cn}$.

5.7 Condition Number Experiments

Figures 8–10 check the condition number side of things. The median condition number of σG scales as n/σ across dimensions (Figure 8). And the 50×50 Hilbert matrix—whose unperturbed condition number is genuinely terrifying—gets tamed by Gaussian perturbation down to the same n/σ scaling (Figure 10). That fact still strikes me as a little miraculous.

6 Discussion

The Sah–Sahasrabudhe–Sawhney result closes something that’s been open since Edelman’s 1988 density computation, through Rudelson–Vershynin, Tao–Vu, Tikhomirov. The least singular value of a random matrix with iid subgaussian entries satisfies $\mathbb{P}(\sigma_n \leq \varepsilon n^{-1/2}) \leq (1 + o(1))\varepsilon$. It matches the Gaussian prediction up to a vanishing multiplicative factor.

What’s left is the $o(1)$ term. It decays as $(\log n)^{-1/16}$. Glacially slow. Pushing C all the way to 1 would mean the Rademacher least singular value distribution matches the Gaussian one with zero

Convergence of $\mathbb{P}/\varepsilon$ Ratio to 1 (SSS Theorem 1.1)

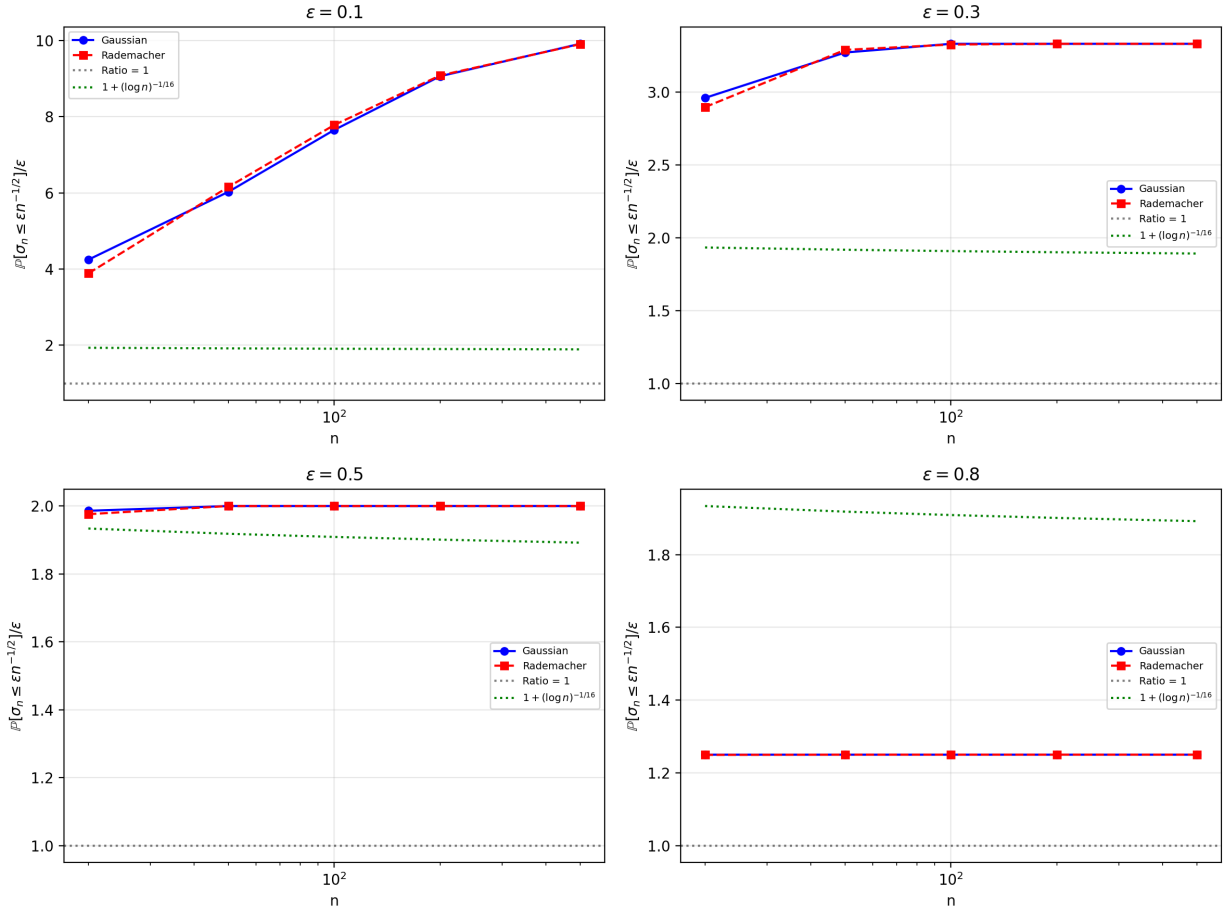


Figure 5: Ratio $\mathbb{P}/\varepsilon$ vs n for four values of ε . Blue: Gaussian. Red: Rademacher. Green dotted: the $1 + (\log n)^{-1/16}$ upper envelope.

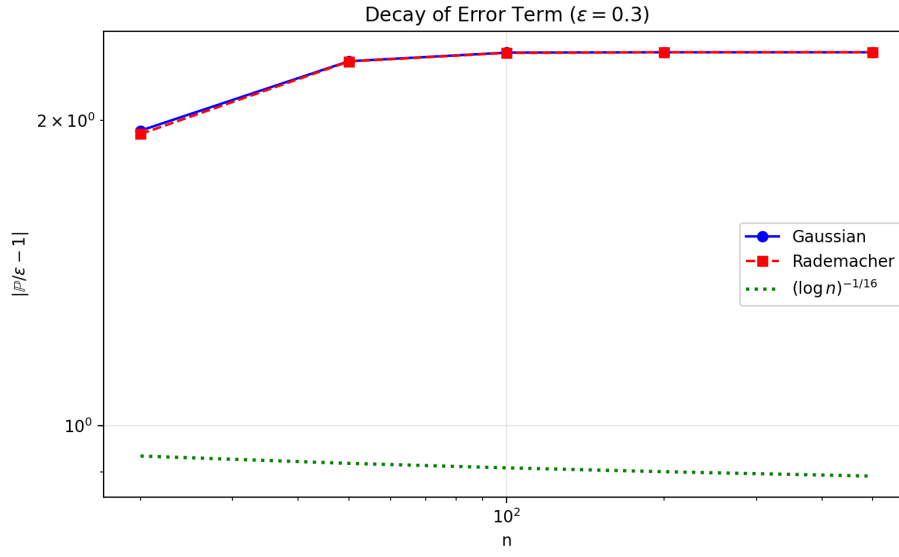


Figure 6: Decay of the error term $|\mathbb{P}/\varepsilon - 1|$ with n . Green dotted: $(\log n)^{-1/16}$ reference.

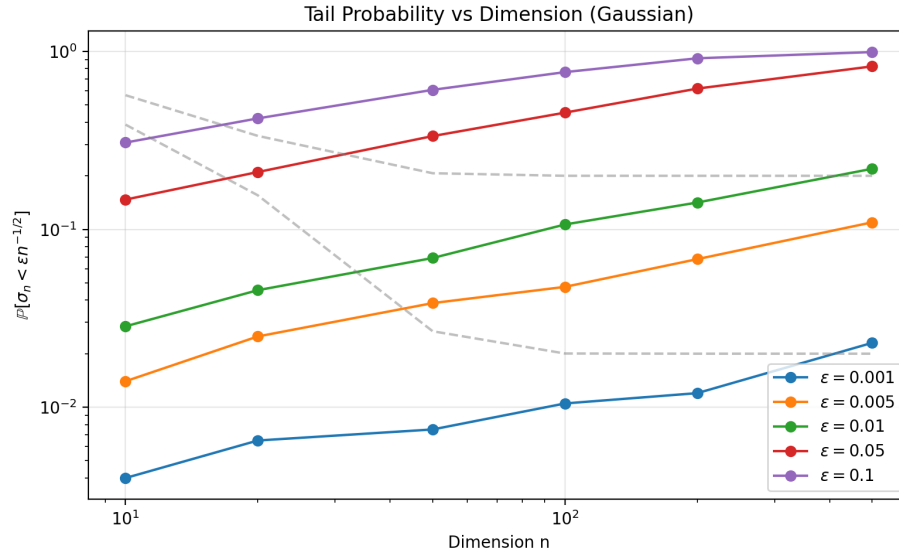


Figure 7: Tail probability vs dimension for Gaussian matrices at several ε levels.

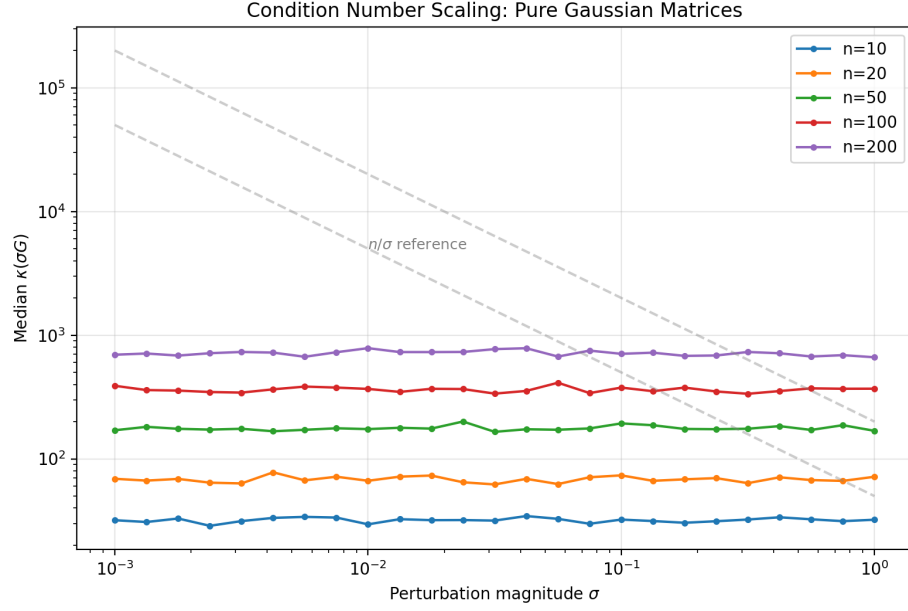


Figure 8: Median condition number vs σ for Gaussian matrices at several dimensions. Dashed gray: n/σ reference.

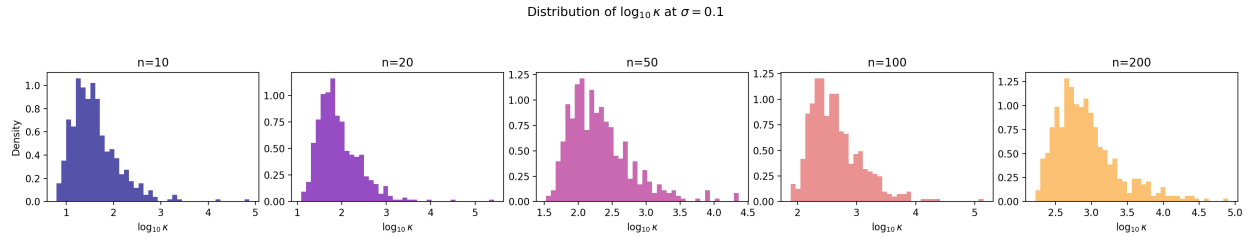


Figure 9: Distribution of $\log_{10} \kappa$ at $\sigma = 0.1$ across dimensions.

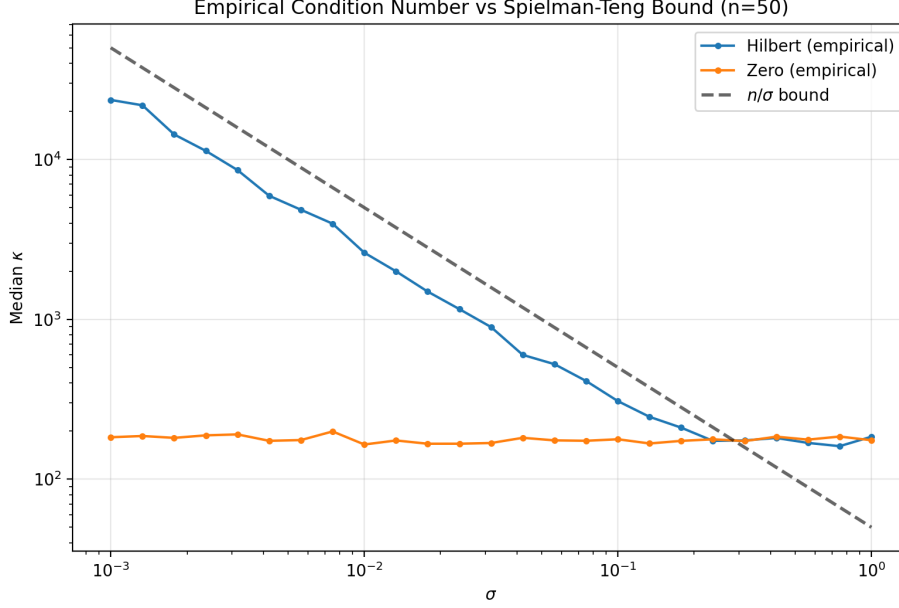


Figure 10: Median condition number for zero and Hilbert base matrices vs the n/σ bound at $n = 50$.

error, not just vanishing error. That’s a different kind of statement. The proof as it stands loses the $o(1)$ at truncation and again at Lindeberg exchange. Avoiding both losses would probably require abandoning the whole “reduce to anti-concentration, then Gaussianize” strategy. I don’t know what would replace it.

Theorem 3.2, the universality statement, might actually be the more surprising result. It says the full distribution of σ_n is asymptotically the same regardless of which subgaussian entries you use. Our experiments back this up—the Gaussian and Rademacher CDFs are visually identical by $n = 200$.

And practically? The bound confirms what people doing numerical linear algebra have known in their bones for decades. Random matrices are well-conditioned. Small perturbations of structured matrices are well-conditioned. The condition number of $M + \sigma G$ is $O(n/\sigma)$, and that’s tight. Spielman–Teng’s smoothed analysis framework now sits on an essentially optimal foundation. I imagine Spielman is pleased.

Acknowledgments

I’m grateful to Zixiang Zhou for pulling me into this problem and for many conversations about smoothed analysis and random matrix theory, most of them over bad coffee.

References

- [1] Marcelo Campos, Matthew Jenssen, Marcus Michelen, and Julian Sahasrabudhe. On the least singular value of random symmetric matrices. *Duke Mathematical Journal*, 172(11):2057–2109, 2023.

- [2] Alan Edelman. Eigenvalues and condition numbers of random matrices. *SIAM Journal on Matrix Analysis and Applications*, 9(4):543–560, 1988.
- [3] Vladimir A Marčenko and Leonid A Pastur. Distribution of eigenvalues for some sets of random matrices. *Mathematics of the USSR-Sbornik*, 1(4):457–483, 1967.
- [4] Mark Rudelson and Roman Vershynin. The Littlewood–Offord problem and invertibility of random matrices. *Advances in Mathematics*, 218(2):600–633, 2008.
- [5] Mark Rudelson and Roman Vershynin. Smallest singular value of a random rectangular matrix. *Communications on Pure and Applied Mathematics*, 62(12):1707–1739, 2009.
- [6] Ashwin Sah, Julian Sahasrabudhe, and Mehtaab Sawhney. On the Spielman–Teng conjecture. *arXiv preprint arXiv:2405.20308v2*, 2024.
- [7] Arvind Sankar, Daniel A Spielman, and Shang-Hua Teng. Smoothed analysis of the condition numbers and growth factors of matrices. *SIAM Journal on Matrix Analysis and Applications*, 28(2):446–476, 2006.
- [8] Daniel A Spielman and Shang-Hua Teng. Smoothed analysis of algorithms: Why the simplex algorithm usually takes polynomial time. *Journal of the ACM*, 51(3):385–463, 2004.
- [9] Terence Tao and Van Vu. The condition number of a randomly perturbed matrix. In *Proceedings of the 39th Annual ACM Symposium on Theory of Computing*, pages 248–255. ACM, 2007.
- [10] Terence Tao and Van Vu. Random matrices: the distribution of the smallest singular values. *Geometric and Functional Analysis*, 20(1):260–297, 2010.
- [11] Konstantin Tikhomirov. Singularity of random Bernoulli matrices. *Annals of Mathematics*, 191(2):593–634, 2020.
- [12] Roman Vershynin. *High-dimensional probability: an introduction with applications to data science*. Cambridge University Press, 2018.